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TRIASSIC PALEOMAGNETIC RESULTS FROM THE YANGTZE BLOCK, S.E. CHINA

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Abstract. In an effort to contribute to the refinement of the APWP for the Yangtze Block, samples for paleomagnetic analyses were recovered from early Triassic sediments in Hunan Province in southeast China. The samples were taken from opposite limbs of a large syncline in order to apply the fold test and rock magnetic experiments were performed to determine the carriers of remanence. However, instead of refining the APWP, the results present some problems.

While the sites pass the fold test, the declinations deviate clockwise with respect to the reference APWP for the block but agree well with other reported results. Due to the paucity of reliable Triassic results from the Yangtze Block and to the small number of sites from this study, it is difficult to determine whether these sites have been rotated or the APWP is incorrect. Both possibilities are explored.

Introduction

Though paleomagnetic work in China is little more than a decade old, studies in southeast China have, in recent years, yielded enough results to construct an Apparent Polar Wander Path (APWP) for the Yangtze Block. The first preliminary APWP for the block was presented by Lin (1984) based on the results of his large-scale paleomagnetic study of China. Since then, more poles for the Yangtze Block have been published, with some of the most reliable results coming from the Mesozoic (e.g. Lin et al., 1985; Opdyke et al., 1986; Heller et al., 1988; Steiner et al., 1989). The most recent and comprehensive APWP for Yangtze from the Permian to the Present, was published by Enkin et al. (1992), incorporating and critically evaluating the results to date.

The problems of remagnetization and working in a tectonically disturbed area, however, have hampered refinement of the APWP for this block. Although the Mesozoic has been studied in perhaps the most detail - the Triassic in particular - there are still few reliable poles reported in the literature. This scarcity of reported poles for south China from the Mesozoic and large statistical ambiguities have resulted in an APWP with large confidence limits in which the trend of the pole path is emerging but the detail is still lacking.

Paleomagnetic studies of two Triassic formations from the Yangtze Block were undertaken with the aim of helping to refine the APWP for the block. The samples were taken at four sites, two each from opposite limbs of a large syncline in Hunan Province so that a fold test could be applied. Folding is Jurassic to Early Cretaceous in age (Hsü et al., 1990).

Sampling Sites and Methodology

The Early Triassic (Induan Stage) Daye Formation was sampled at two sites - one from each limb of a large NE-SW trending syncline on the Yangtze Block near the village of Hongjiaguan (110°E, 29.5°N) (Figure 1). The Daye Formation consists of medium-grey limestones and blue-grey marls with intercalated mudstone. Bivalves and ammonids are the dominant taxa. Eighteen cores from both the limestones and marls were taken for this study.

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The Triassic Jialingjiang Formation overlies the Daye Formation and was sampled from both limbs of the same syncline. While the age of the Jialingjiang is thought to be Middle Triassic by many Chinese geologists, it is also reported to span from the Early Triassic into the Middle Triassic Anisian stage (Yang et al., 1986). The Jialingjiang is made up mostly of limestones, dolomitic limestone and brecciated dolomites. Bivalves, ammonoids, conodonts and foraminifera are seen in these rocks.

All samples were subjected to either stepwise thermal demagnetization using a Schonstedt TSD-1 Thermal Specimen Demagnetizer or A.F. demagnetization using an A.F. demagnetizer constructed at the ETH. The remanence was measured with a 2G 3-axis SQUID magnetometer interfaced with a Hewlett-Packard HP-1000 computer. The magnetic vectors were analyzed using standard paleomagnetic techniques (Zijderveld, 1967; Kirschvink, 1980; Fisher, 1953). IRM experiments were performed using an Oxford Instruments electromagnet generating peak D.C. fields of 1T.

Paleomagnetic Results

The Daye Formation

The NRM intensities of the Daye samples varied little and had a mean intensity of 7.45 x 10^{-4} A/m. The NRM directions are all normal polarity with only two exceptions but do not appear to be representative of a recent overprint. The samples from the southeast limb were taken from a grey marl unit and had slightly higher blocking temperatures than the

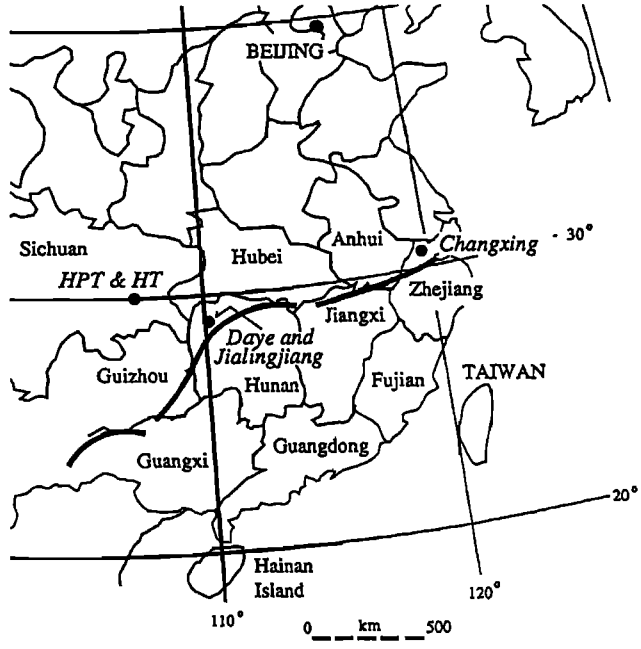


Fig. 1. Location map showing the sampled area as well as the location of the sampling sites of Steiner et al. (1989) (HP and HPT) and the Changxing P-Tr section. The heavy dashed line is the suture zone between the Huanan and Yangtze Blocks (Hsü et al., 1990).

samples from the limestone unit of the northwest limb. Nearly all samples exhibited a stable - usually two-component - decay to the origin at temperatures between 375°C and 500°C (Figure 2a). In two of the limestone samples, a large decrease in intensity between 100°C and 125°C is seen, probably indicating the presence of goethite. Most of these samples had low blocking temperatures on the order of 350°C to 400°C. Isothermal remanent magnetization (IRM) acquisition experiments reveal a dominance of low coercivity minerals and the IRM becomes unblocked upon thermal demagnetization to temperatures between 525°C and 575°C (Figure 2c). This indicates that magnetite is the likely carrier of remanence in these samples.

Equal area projections of the stable, high-temperature directions show that the sites pass a fold test at the 99% confidence level and both normal and reversed polarities are seen (Figure 3a). The mean direction is, however, not the expected Early Triassic direction for the Yangtze Block (Enkin et al., 1992). The pole position for these sites is 205.1°E and 39.8°N, which is in close agreement with Early Triassic results obtained from two of the four formations (HPT & HT) studied by Steiner et al. (1989) in Sichuan Province. Their results, however, are displaced from the APWP for the Yangtze Block and are thought to be rotated approximately 15° clockwise (Figure 4). They report that the sense of this rotation is supported by structural data but the amount of rotation can only be postulated from the paleomagnetic evidence (Steiner et al., 1989).

Considering the agreement with the results of Steiner and others (1989) and the fact that these sites pass a fold test, it appears that these sediments may preserve a primary magnetization acquired during the Early Triassic but the sites have been subsequently rotated with respect to the Yangtze Block APWP. It is also possible that these sites (HPT, HT, and Daye) preserve the true Early Triassic pole position for the block and the APWP is in error.

The Jialingjiang Formation

The NRM intensities of the Jialingjiang Formation are closely grouped around a geometric mean of 1.07 x 10^{-3} A/m. The NRM directions were fairly scattered and do not appear to be related to the Brunhes field. Samples from both sites were demagnetized using thermal and A.F. treatments which were equally effective at resolving a stable direction. Low blocking temperatures - between 400°C and 525°C - are seen in all of the samples (Figure 2b). The effectiveness of A.F. treatments indicates that the samples are likely dominated by low coercivity minerals.

IRM acquisition curves also indicate that the remanence is dominated by low coercivity minerals with virtually no contribution from minerals with high coercivities (Figure 2d). The samples are saturated by 0.2T to 0.3T which is slightly high for magnetite but may indicate the presence of small amounts pyrrhotite. Unblocking temperatures between 500°C and 575°C are seen in the thermal demagnetization of IRM. This evidence shows that the stable, high-temperature directions are probably carried by magnetite.

As seen in the equal area projections of the stable directions, this formation also passes the fold test at the 99% confidence level (Figure 3b). The paleopole position calculated for these sites is 191.0°E and 34.0°N, and it appears again, that these sites have been rotated in the same sense as the Daye samples and the results of Steiner and others (1989) with respect to the reference APWP (Figure 4). This pole and the Daye pole may actually define a rotated APWP segment spanning the Early Triassic.

Discussion and Conclusions

Results from the Early Triassic Daye and early to Middle Triassic Jialingjiang Formations present evidence that these sites may have been rotated in a manner similar to two

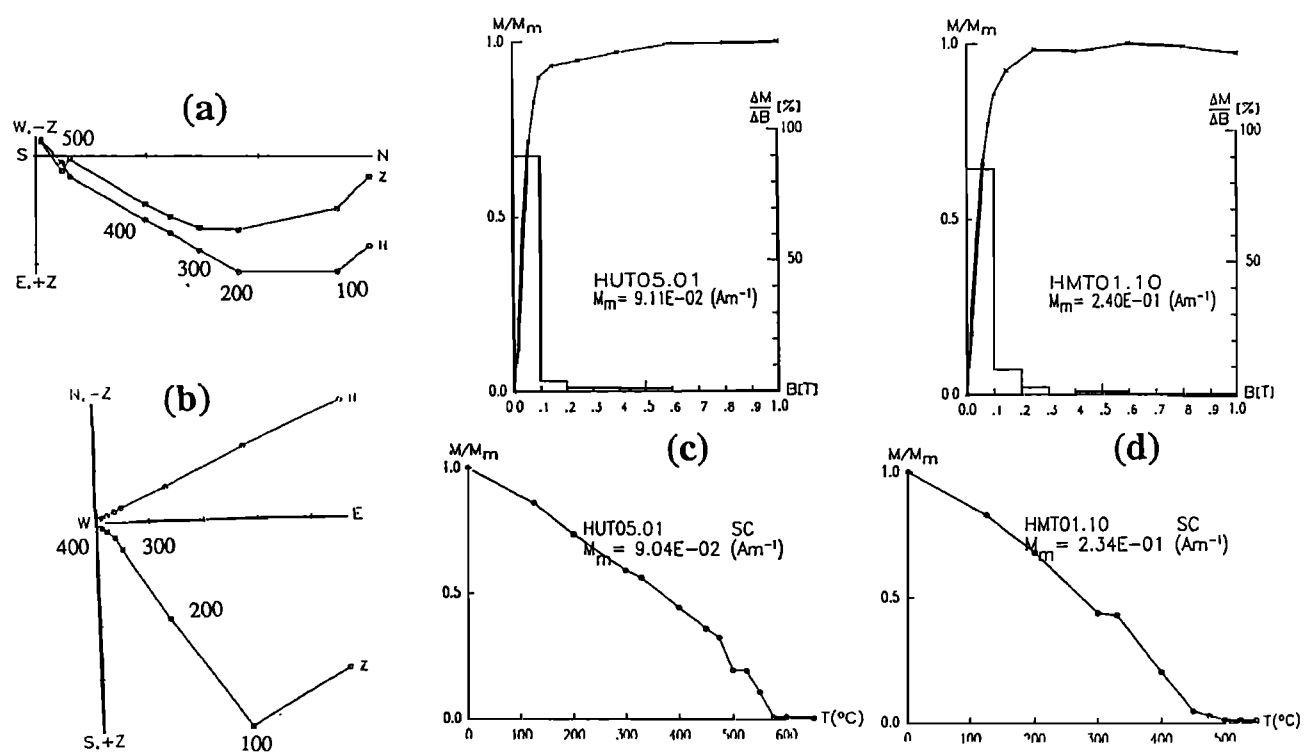


Fig. 2. (a) Representative Zijderveld diagram for the Daye Formation. (b) Representative Zijderveld diagram for the Jianlingjiang Formation (c) IRM acquisition and thermal demagnetization for the Daye Formation (d) IRM acquisition and thermal demagnetization for the Jialingjiang Formation.

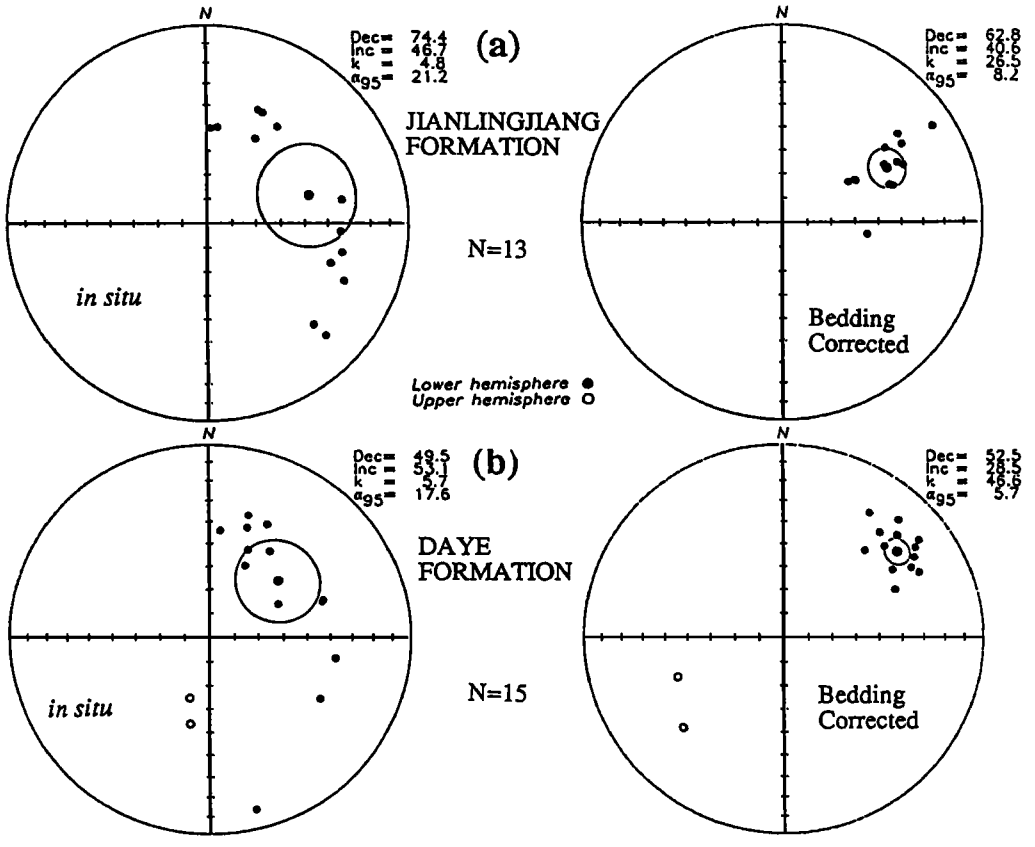


Fig. 3. (a) Equal area projections of the high-temperature directions from the Jianlingjiang Formation (b) Equal area projections of the high-temperature directions from the Daye Formation.

formations studied by Steiner et al. (1989). Both results pass a fold test, as do the sites of Steiner et al., and these data may have implications for large scale rotations in the region. Though both formation means appear to be rotated in the same sense and by a similar amount as the results of Steiner et al.,

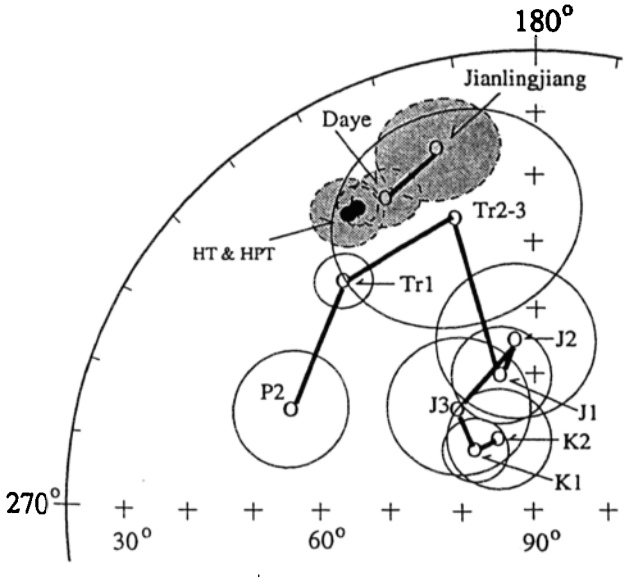


Fig. 4. APWP of Enkin et al. (1992) and the location of the rotated poles of this study (open circles, light grey dashed) with A95 circles of confidence and Steiner et al. (1989) (solid circles, light grey dashed).

the Daye pole falls within the confidence limits of the crudely determined middle to late Triassic pole of Enkin et al. (1992) (Table 1). This could be an indication that the Daye Formation has been remagnetized during this time. The close grouping of the Daye pole with the poles from Steiner et al., however, does appear to present a stronger case for rotation of these sites.

The close grouping of the data from this study - especially the Daye Formation - with the poles of Steiner et al. implies that the sites from both studies may have been rotated as a single unit. The close proximity of the HPT and HT sites from the Steiner et al. study with apparently unrotated sites from the same study may, however, indicates that the rotations, though in the same sense, could also be local thrust sheet rotations.

This being the case, it seems that a fairly large region of the Yangtze Block may have been affected by these rotations. This area would include the area sampled by Steiner and others and western Hunan Province, from where the samples from this study come. If these two study areas were rotated as a coherent unit, then the rotation would also have to include the region between the study areas - comprising parts of extreme northeastern Guizhou, southwestern Hubei and northwestern Hunan Provinces.

This is not, however, the only interpretation of these data. It is also possible that the results of this study and Steiner et al. are representative of the stable Yangtze Block during the early to Middle Triassic and that the APWP of Enkin et al. (1992) is in error. This interpretation would imply that the Triassic results used by Enkin et al. to calculate mean Triassic poles are themselves rotated with respect to the rest of the block.

The results used by Enkin et al. in their determination of the early and Middle Triassic poles for the Yangtze Block come almost exclusively from Sichuan Province. One result

TABLE 1. Comparison of Triassic results from this study with the results of Steiner et al. (1989) and the Triassic directions from the Yangtze Block (Enkin et al., 1992).

NAME	AGE	N	Pole LAT	Pole LON	A95
Daye	Tr_1	15*	39.8	205.1	5.7
Jianlingjiang	Tr_{1-2}	14*	34.0	191.1	8.2
HPT (Steiner et al.)	Tr_1	62*	38.5	209.8	3.2
HT (Steiner et al.)	Tr_1	26*	38.7	211.8	6.0
Reference poles	Tr_1	10**	45.7	217.7	4.5
from Enkin et al.	Tr_{2-3}	5**	45.1	194.4	16.8

* = number of samples, ** = number of sites

from Opdyke et al. (1986) comes from the Guizhou-Sichuan border. The question must then be raised whether these results are representative of the Yangtze Block outside of that limited area. To answer this question we must look to the one result used by Enkin et al. in the calculation of the Early Triassic pole from the eastern part of the block at the Changxing Permian-Triassic type section in Zhejiang Province (Li and Wang, 1987). Those results are in good agreement with the reference APWP and the rock magnetic properties have been studied and the results have been confirmed by other workers (Dobson et al., 1992).

In view of these facts, the Changxing results appear to confirm that the Early Triassic pole of Enkin et al. is representative of the Yangtze Block and the poles reported in this study and Steiner et al. have been rotated. In their paper, however, Enkin et al. use the Steiner poles (HT and HPT) in their calculations of a mean Early Triassic paleopole. Since the Steiner poles appear not to be representative of the unrotated block, they should not be used in calculations of an Early Triassic pole for the Yangtze Block.

Though the amount and sense of rotation seen in this study is similar to the data from Steiner et al., more results are needed to determine whether both sampling areas rotated as a single unit or were affected by local rotations of thrust sheets in the same sense. More work outside of Sichuan Province is needed as well, in order to recognize the extent of thrust sheet rotations within the block and to further refine the APWP.

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